

A Field Guide to Paper

Grain, weight, and how a sheet behaves under pressure

QUIRE PRESS

Letterpress and bindery, established in a damp room

First edition. Printed for the bench, not the shelf.
Six sections on grain, weight, sizing and the fold.

Set in Times and Helvetica. Composed and printed in house.

Reading a Sheet

Paper is not a neutral surface. It arrives at the press carrying a history: the direction the fibres travelled on the wire, the amount of size stirred into the pulp, the humidity of the room where it was cured and the room where it now waits. Learn to read those things with your hands and half the trouble of a press run is over before you ink the first form.

GRAIN DIRECTION

Every machine made sheet has a grain. As the pulp runs onto the moving wire, the fibres line up with the direction of travel, the way reeds lean downstream. Once the water is gone that alignment is permanent. The sheet is stiffer across the grain and more willing along it, and every later decision, folding, printing, gluing, curling, is an argument with that one fact.

There are three quick tests and none of them requires a tool. Tear a narrow strip from a scrap: along the grain the tear runs long and straight, across it wanders and feathers. Lay the sheet over the edge of the bench in both directions and feel which way it droops more readily, because that is the grain. Or moisten one side and wait: the sheet curls with its axis parallel to the grain, since the fibres swell in girth far more than in length.

Naming the direction

Stock is sold as long grain or short grain, and the words describe the sheet, not the press. Long grain means the fibres run parallel to the long side of the sheet as it is cut. When you order paper for a book, order it so the grain will end up parallel to the spine after the last fold, and say so on the order, because the mill cannot guess at your imposition.

WHY THE FOLD CARES

A fold made along the grain slides the fibres past one another. A fold made across the grain snaps them, and the crease shows it: a chalky line where the coating or the surface has fractured, ragged under a loupe and unforgiving on a dark ink. On heavy stock the difference is not subtle. Score first, always, and score with a rounded rule rather than a sharp one, so the fibres are pushed aside instead of cut.

A book folded against the grain will never lie flat. The pages fight the spine, the fore edge fans, and the whole block develops a gentle dome within a season. It is the single most common fault in an otherwise careful piece of work, and it is invisible until it is permanent.

A HABIT WORTH KEEPING

Mark the grain on every ream the moment it comes off the pallet. A pencil arrow on the wrapper, drawn along the grain, costs three seconds and settles a dozen later arguments. Mark the offcuts too, because a quarter sheet with no wrapper is a sheet with no history, and by the time you are testing tears with your thumbnail at the guillotine you have already lost the afternoon.

Weight, and the Two Ways We Measure It

Two numbers are printed on every wrapper and they measure the same sheet in two unrelated ways. Knowing which one is being quoted saves an expensive order.

GRAMMAGE

Grammage is honest. It is the mass in grams of one square metre of the stock, whatever size that stock happens to be cut to, and it is written gsm. A 120 gsm sheet is 120 gsm in any format, so two papers can be compared without arithmetic. Nothing about the number tells you how thick the sheet feels, though. A soft, bulky cotton at 175 gsm can stand taller than a dense coated stock at 250.

BASIS WEIGHT

Basis weight is a survival from the trade. It is the weight in pounds of a ream, five hundred sheets, of the paper in its own basic size, and the basic size differs by grade. Text papers are weighed at 25 by 38 inches, cover papers at 20 by 26, bristol at 22 and a half by 28 and a half. So 80 lb text and 80 lb cover are not remotely the same object, and the grade word after the number is doing all the work.

Two rules of thumb see you through most conversations. To go from text basis weight to grammage, multiply by about one and a half. To go from cover, multiply by about two and seven tenths. Neither is exact, and neither needs to be, because the mill prints both figures on the wrapper.

SIZING

Sizing is the glue worked into or onto the sheet to slow the passage of liquid. Unsized paper is blotter: ink sinks, spreads, and goes dull. Heavily sized paper holds ink on the surface, which keeps a fine serif crisp but leaves it sitting wet for longer. Internal sizing is beaten into the pulp, surface sizing is a tub bath afterwards, and printmaking papers usually have both.

For letterpress a moderately sized, softly bulked sheet is the friend of a deep impression. It takes the punch of the type without shattering and gives that quiet, tactile bruise on the back of the leaf that people reach out to touch

without knowing why.

DECKLE EDGES

The deckle is the frame that fences the pulp on the mould. Where the slurry thins against it, the sheet ends in a soft feathered edge rather than a cut one. A true deckle is a record of how the sheet was made. Two deckles and two cut edges mean a mould made sheet; four deckles usually mean it was made by hand, one pull at a time.

A torn edge can be coaxied to imitate a deckle with a wetted line and a steel straightedge, but the fibres will tell on you under raking light.

HOW A SHEET MOVES

Paper is a sponge with opinions. Cellulose fibres take water out of the air and give it back, and they swell across their width far more than along their length. So a sheet gains and loses size mostly across the grain: a wide sheet can move a good two millimetres between a wet morning and a dry afternoon, while the same sheet barely moves at all in the other direction.

That is why register drifts on a humid day and why the second colour never quite lands on the first if the pile has been left uncovered overnight. It is also why a stack that has been leaning against a cold outside wall develops wavy edges: the rim of the pile has drunk from the room while the middle has not, and the sheet ends up longer at its edges than through its centre.

The remedy is patience rather than cleverness. Let stock stand in its wrapper in the pressroom for two days before you cut it, keep the pile covered between passes, and print a multi colour job in one sitting if the weather is turning. If the room swings more than ten points of relative humidity across a day, a cheap hygrometer over the bench will earn its price in the first season.

THREE

A Working Table of Stocks

These are the seven papers kept on the racks by the guillotine. The grammage is the mill figure; the basis weight is rounded to whatever the trade would call it. Caliper is not listed because it drifts with the weather, and because a micrometer settles the question faster than a table can.

STOCK	GRAMMAGE	BASIS WEIGHT	WHAT WE REACH FOR IT
Glassine tissue	45 gsm	12 lb text	Interleaving wet prints
Laid text, cream	118 gsm	32 lb text	Text pages, light forms
Wove text, natural	148 gsm	40 lb text	Long runs, tight register
Cotton rag, soft	175 gsm	47 lb text	Deep impression, few colours
Mould made, deckled	250 gsm	92 lb text	Broadsides and menus
Dyed cover	300 gsm	111 lb cover	Wrappers and slipcases
Binder board, grey	590 gsm	218 lb cover	Cased boards, boxes

A stock two steps heavier than you think you need will usually print better and fold worse. When the piece has to fold, choose the grain first and the weight second.

CHOOSING FOR THE RUN

Start from the ink, not the paper. A solid, an even tint or a halftone wants a hard, well sized, smooth surface that keeps the film sitting on top; anything textured will break the solid into a field of tiny holidays. Type, rules and line work want the opposite: a softer, bulkier sheet that lets the face bed in and gives that quiet shoulder around every letter.

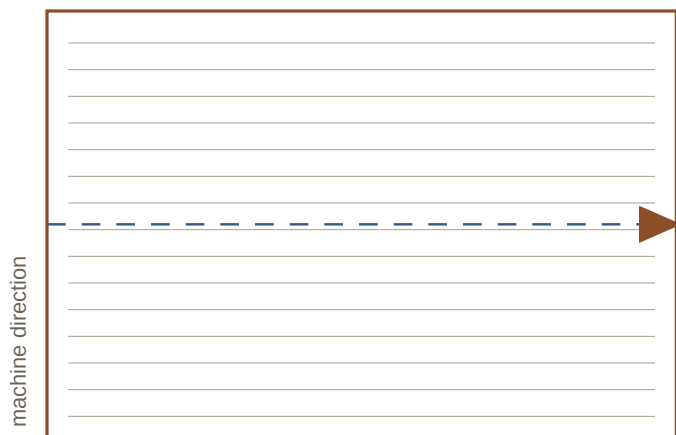
Then count the passes. Every colour is another trip through the press and another chance for the sheet to change size, so a four colour job argues for a heavier, more stable stock than a single colour job of the same size. Add ten percent to the order for make ready on a short run and five on a long one, and add more than that if the stock is a special order the mill will not repeat.

Finally, hold the finished thing in your head. A menu is picked up wet handed twice an evening and wants a hard, dyed through cover. A book of poems is read once and reread for thirty years and wants a soft sheet that opens flat. The right stock is the one that is still doing its job when nobody is thinking about it.

Grain and the Folding Scheme

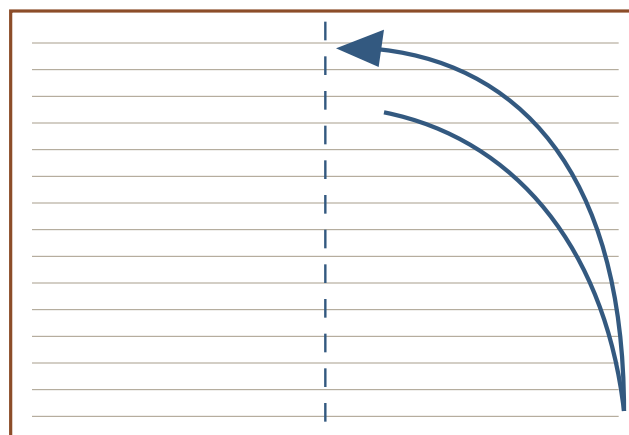
The diagram below shows a flat sheet with its grain running left to right, and the two folds that turn it into a four leaf section. The first fold runs with the grain and behaves. The second crosses it and must be scored.

FLAT SHEET, GRAIN LEFT TO RIGHT



Fold one runs with the grain. It slides, and needs no score.

SECOND FOLD, ACROSS THE GRAIN



Fold two snaps the fibres. Score it, or accept a chalky crease.

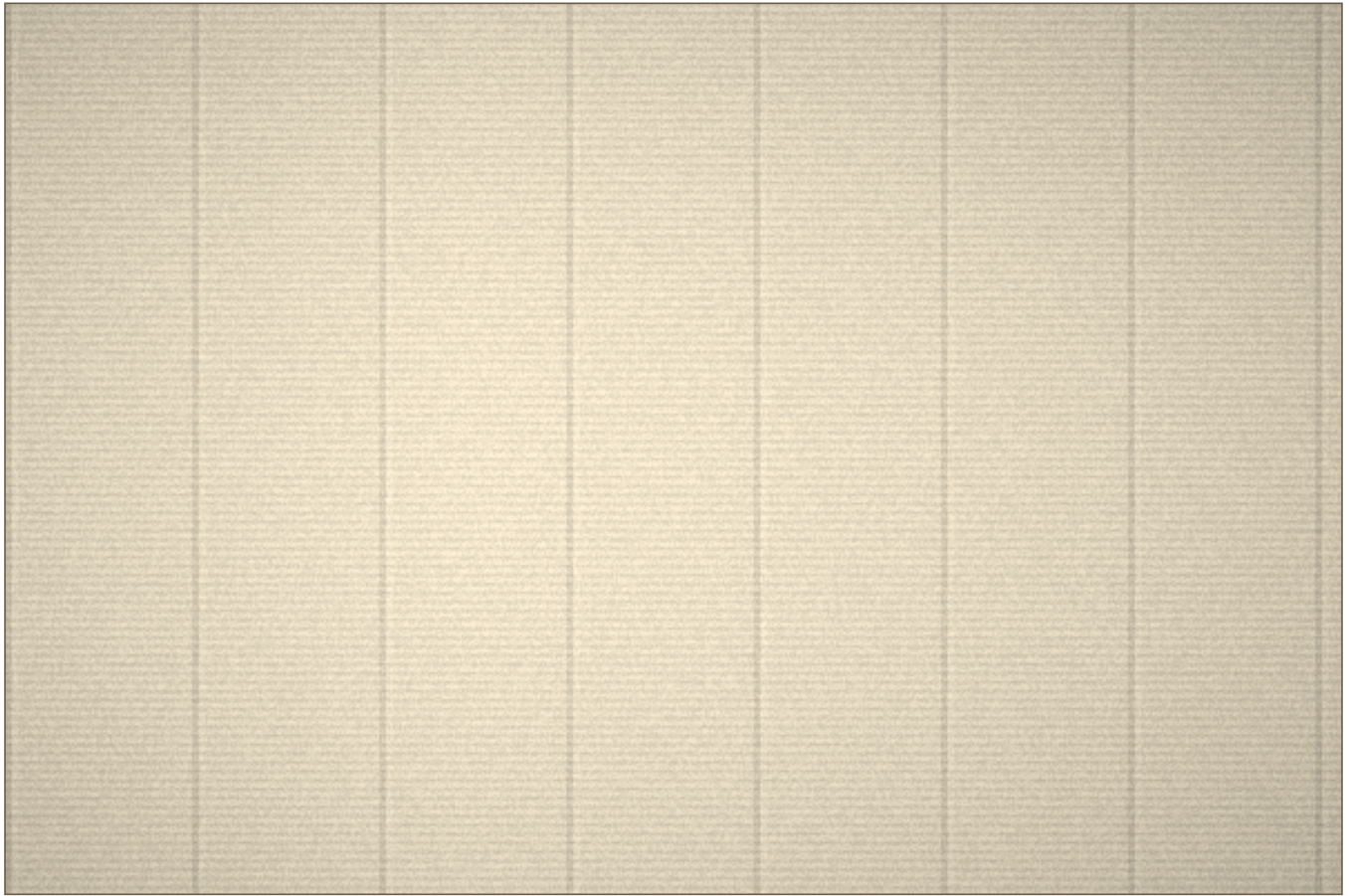
Impose so that the last fold, the one nearest the spine, runs with the grain. If the sheet will not allow it, turn the imposition rather than the paper: it is cheaper to reset a form than to explain a fanned fore edge to somebody who paid for a book. Score at a depth of roughly a third of the caliper, on the outside of the fold, and let the sheet rest an hour in the room it will be folded in before you touch it.

THE SECTION IN FOUR LEAVES

- Cut the sheet so the grain runs the long way, parallel to what will become the spine.
- Score the second fold now, while the sheet is flat and square to the bench, not later when it is already folded once.
- Make the first fold along the grain, bone it down from the middle outwards, and let it rest.
- Make the second fold across the scored line, then trim the fore edge last, after the section has been under a weight overnight.

Nothing on this page survives contact with a sheet that has been cut the wrong way round, so check the grain before the guillotine and not after it.

The Look of a Laid Sheet



Raking light across a mould made laid sheet. The close ruled shadows are the laid wires; the wider bands running the other way are the chain lines that stitch the mould together.

Hold a laid sheet up to a window and the mould is still there in it. The pattern is not printed and not embossed: it is a thickness difference, laid down while the sheet was still water, and it survives pressing, drying and a century of shelving. On a good sheet the wires sit about a millimetre apart and the chains about thirty. Some machine made papers imitate the effect with a dandy roll, and the imitation is honest enough, but the roll repeats and the mould does not.

For our purposes the practical question is which way the wires run, because the laid pattern is usually laid across the grain on a mould made sheet. That makes it a useful second opinion when the tear test is ambiguous, and it makes the sheet a nuisance for solids, since the ink will sit a little heavier in every hollow. Print type on it, print rules on it, and leave the flat tints to a wove.

Buy laid stock by the mould mark if you can. Two mills making the same nominal weight will space their wires differently, and once a reader has your first book in hand the second one ought to feel like it came from the same shop. Keep a labelled sheet of every stock you have used pinned above the bench, with the date and the supplier written along the deckle in pencil. It is the cheapest reference library a press can own, and the only one that answers a question by being held up to the light.